

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 22

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Recap of Previous Lecture



- ✓ Topic Monoid
- ✓ Topic Group
- ✓ Topic Abelian group / Commutative group
- ✓ Topic Practice Questions
- ✓ Topic Finite group



Topics to be Covered



Topic

Finite group

Topic

Addition modulo 'm' $\oplus_m$

Topic

Multiplication modulo 'm' $\otimes_m$

Topic

Order of an element in a group $(G, *)$

Topic

Subgroup

H.W



#Q. Which of the following statement is/are not true.

A

$\{0, \pm 2k, \pm 4k, \pm 6k, \dots\}$ is a group with respect to addition where k is any fixed positive integer (True) ✓

B

$\{x \mid x \text{ is real number and } 0 < x \leq 1\}$ is a group with respect to multiplication ✓
it is a monoid but not a group $\text{inv}(0.5) = 2 \notin \text{Set} \therefore \text{Not a group}$

C

$\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication
group ✓ (True) $2^{n_1} * 2^{n_2} = (2)^{n_1+n_2} \therefore \text{Closed}$

D

None of these

$2^0 \in \text{Set} \therefore \text{identity exists}$



Topic : Finite Group

Cube roots of unity are $1, \omega, \omega^2$

$$\omega^3 = 1$$

The set $\{1, \omega, \omega^2\}$ form a finite group of order = 3
w.r.t. multiplication.

- ① Closed ✓
- ② Associative ✓
- ③ Identity is present ✓
- ④ Inverse ✓

To check for inverse
We can use Composition
table

Multiplication

	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	$\omega^3 = 1 = e$
ω^2	ω^2	$\omega^3 = 1 = e$	$\omega^4 = \omega$



Topic : Finite Group

Four roots of unity are $1, -1, i, -i$
 $i^2 = -1$ & $i^4 = 1$



The set $\{1, -1, i, -i\}$ form a finite group of order = 4
w.r.t. multiplication.

- ① Closed ✓
- ② Associative ✓
- ③ Identity is present ✓
- ④ Inverse ✓

$$\begin{aligned} \text{inv}(1) &= 1 \\ \text{inv}(-1) &= -1 \\ \text{inv}(i) &= -i \\ \text{inv}(-i) &= i \end{aligned}$$

	1	-1	i	-i
1	$1=e$	-1	i	-i
-1	-1	$1=e$	-i	i
i	i	-i	$i^2 = -1$	$-i^2 = -(-1) = 1=e$
-i	-i	i	$-i^2 = 1=e$	$i^2 = -1$

A set of 'n' roots of unity will form a finite group
of order = n w.r.t. multiplication



Topic : Addition modulo 'm' $\oplus_m$

it gives remainder

Let m is any fixed positive integer
for any two non-negative integers a & b ,

$$a \oplus_m b = \begin{cases} (a+b) & , \text{ if } (a+b) < m \end{cases}$$

|||
 $(a+b) \bmod m$

r

otherwise

Where ' r ' is the remainder
obtained on dividing $(a+b)$ by m

$$\begin{aligned} 2 \oplus_6 3 &= 5 \\ 4 \oplus_6 5 &= 3 \end{aligned}$$



Topic : Multiplication modulo 'm' $\otimes_m$

Let m be any fixed positive integer,
for any two non negative integers a & b ,

$$a \otimes_m b = \begin{cases} (a.b) & \text{if } a.b < m \\ \pi & \text{otherwise} \end{cases}$$

$\equiv$
 $(a.b) \text{ Mod } m$

otherwise

Where π is the remainder
Obtained on dividing $(a.b)$ by m .

$$\begin{aligned} 2 \otimes_7 3 &= 6 \\ 4 \otimes_7 5 &= 6 \end{aligned}$$



Topic : NOTE

① The set $\{0, 1, 2, \dots, (n-1)\}$ form a finite group of "order = n " w.r.t. binary opⁿ $\oplus_n$.

eg. The set $\{0, 1, 2, 3, 4\}$ form a finite group of order = 5, w.r.t. $\oplus_5$.

$$\begin{aligned} \text{inv}(0) &= 0 \\ \text{inv}(1) &= 5-1=4 \\ &\Rightarrow 1 \oplus_5 4 = 0 \end{aligned}$$

$$\begin{aligned} \text{inv}(2) &= 5-2=3 \\ &\Rightarrow 2 \oplus_5 3 = 0 \end{aligned}$$

$$\text{inv}(3) = 2$$

$$\text{inv}(4) = 1$$

① Set is closed w.r.t. $\oplus_5$

② $\oplus_n$ is associative for any +ve integer 'n'

③ Identity element w.r.t. $\oplus_n$ is always '0'.

④ Inverse: $\text{inv}(0) = 0$ {because '0' is identity}

for any other element $x \in \text{Set}$
 $\text{inv}(x) = (n-x)$ w.r.t. $\oplus_n$

Note:-

Let

$$A = \{x \mid x \in \mathbb{N} \text{ and } x < n \text{ and } \text{GCD}(x, n) = 1\}$$

Natural numbers given fixed positive integer

$\rightarrow$ $\left\{ \begin{array}{l} A \text{ is a set of all natural numbers} \\ \text{that are less than 'n' and Coprime to 'n'} \end{array} \right\}$

let $n = 15$,

then

$$A = \{1, 2, 4, 7, 8, 11, 13, 14\}$$

Set of all natural numbers which are less than 15 & Coprime to 15.

If GCD of x & n is '1' then x & n are called Co-prime numbers (or) relatively prime numbers



Topic : NOTE



② The set of all natural numbers that are less than n and Co-prime to n , form a group w.r.t. $\otimes_n$

eg: Let $n=8$, then

$\{1, 3, 5, 7\}$ will form a finite group of order = 4 w.r.t. binary opn $\otimes_8$.

Set of all natural nos that are less than 8, & Co-prime to 8.

→ $\{1, 3, 5, 7\}$ is a finite group of order = 4,
w.r.t. $\otimes_8$

① Closed. ✓

② $\otimes_n$ is associative
for any +ve integer 'n'. ✓

③ '1' is the identity element
w.r.t. $\otimes_n$

④ Inverse:
 $\text{inv}(1) = 1$
 $\text{inv}(3) = 3$
 $\text{inv}(5) = 5$
 $\text{inv}(7) = 7$

$$1 \otimes_8 1 = 1 \text{ (identity)}$$

$$3 \otimes_8 3 = 1 \text{ (identity)}$$

$$5 \otimes_8 5 = 1 \text{ (identity)}$$

$$7 \otimes_8 7 = 1 \text{ (identity)}$$

it just happened that each element
is inverse of itself, it is not
necessary for all such groups.

eg. Let $n=15$, then
 $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is a finite group of order = 8 w.r.t. $\otimes_{15}$

Find inverse of every element of the group.

identity element = 1

$$\text{inv}(1) = 1$$

$$\text{inv}(2) = 8 \Rightarrow \text{inv}(8) = 2$$

$$\text{inv}(4) = 4$$

$$\text{inv}(7) = 13 \Rightarrow \text{inv}(13) = 7$$

$$\text{inv}(11) = 11$$

$$\text{inv}(14) = 14$$

Note: ① $\{1, 3, 5, 7\}$ is a group w.r.t. $\otimes_8$

② $\{1, 3\}$ is also a group w.r.t. $\otimes_8$

③ $\{1, 5\}$ is also a group w.r.t. $\otimes_8$

④ $\{1, 7\}$ is also a group w.r.t. $\otimes_8$

→ Some subset of original group may also be group

⑤ $\{1, 3, 5\}$ is not a group w.r.t. $\otimes_8$

→ i.e. Every subset need not be a group



Topic : NOTE

③ If p is a prime number, then
 $\{1, 2, 3, \dots, p-1\}$ will form a group of
order $= (p-1)$ w.r.t. binary opⁿ $\otimes_p$

eg. $\{1, 2, 3, 4, 5, 6\}$ is a group of
order $= 6$, w.r.t. opⁿ $\otimes_7$

Any no. less than p
will be Co-prime to p ,
 $\therefore \{1, 2, 3, \dots, p-1\}$ is the
set of all natural numbers
that are less than p and
Co-prime to p .

Q:- Which of the following is/are not true.

~~(a)~~ $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\oplus_6$
not closed $2 \oplus_6 4 = 0 \notin \text{Set}$

~~(b)~~ $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\otimes_6$
 $2 \otimes_6 3 = 0 \notin \text{Set} \therefore$ not closed

(True) (c) $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$

~~(d)~~ $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\otimes_5$
 $\hookrightarrow \text{inv}(0) = \text{does not exist w.r.t. } \otimes_5$



Topic : Order of an element in a group $(G, *)$

Let $(G, *)$ be a finite group of order $= |G|$.

For an element $a \in G$,

Order of element 'a' can be denoted by $O(a)$ (or $\langle a \rangle$)
and it is defined as the least positive integer n

such that

$$(a)^n = e \text{ (identity element)}$$

|||

$$\underbrace{a * a * a * \dots * a}_{n \text{ times 'a'}}$$

→ i.e., it can not be '0'

Note:

If 'e' is the identity element of the group,
then

$$(e)^1 = e \text{ (identity)}$$

∴ Order of identity element of the group
is always '1'.

eg. $\{1, -1\}$ is a finite group of order = 2 w.r.t. multiplication
find the order of every element of the group.

$$\bullet \quad 1 = \text{identity} \quad \therefore O(1) = 1$$

$$O(G) = 2$$

$$\bullet \quad (-1)^1 = -1$$

$$(-1)^2 = 1 = \text{identity} \quad \therefore O(-1) = 2$$

eg. $\{1, \omega, \omega^2\}$ is a finite group of order=3 wrt. multiplication
find the order of every element of the group.

• $1 = \text{identity} \therefore O(1) = 1$

→ $(\omega)^1 = \omega \neq 1$

$(\omega)^2 = \omega^2 \neq 1$

$(\omega)^3 = \omega^3 = 1 = \text{identity} \therefore O(\omega) = 3$

→ $(\omega^2)^1 = \omega^2 \neq 1$

$(\omega^2)^2 = \omega^4 = \omega \neq 1$

$(\omega^2)^3 = \omega^6 = 1 = \text{identity} \therefore O(\omega^2) = 3$

$O(G) = 3$

eg. $\{1, -1, i, -i\}$ is a finite group of order = 4 wrt. multiplication
find the order of every element of the group.

1. $1 = \text{identity} \therefore O(1) = 1$

2. $(-1)^1 = -1 \neq 1$

$(-1)^2 = 1 = \text{identity} \therefore O(-1) = 2$

3. $(i)^1 = i$

$(i)^2 = i^2 = -1$

$(i)^3 = -i$

$(i)^4 = 1 = \text{identity} \therefore O(i) = 4$

$O(G) = 4$

4. $(-i)^1 = -i$

$(-i)^2 = -1$

$(-i)^3 = i$

$(-i)^4 = 1 = \text{identity} \therefore O(-i) = 4$



Topic : Properties w.r.t. Finite Group

- ① order of identity element is always '1'.
- ② For any finite group $(G, *)$.
Order of any element of group $(G, *)$ is less than or equal to the order of the group.

Lagrange's
theorem

(And)
Order of each element of group $(G, *)$ divides the order of group $(G, *)$

- ③ order of an element $a \in$ group, will be exactly same as order of its inverse i.e., $O(a) = O(a^{-1})$



Topic : Properties w.r.t. Finite Group

④ Order of an element 'a' of the group can be '1' if and only if element 'a' is the identity element of the group.

⑤ For an element $a \in G$
if $o(a) = 2$ { i.e. $a \times a = e(\text{identity})$ }
then element 'a' is inverse of itself.

H.W.

Q. We know $\{0, 1, 2, 3, 4\}$ is a finite group of
Order = 5, w.r.t. binary opⁿ $\oplus_5$.

Find order of each element of the group.

Order of group is = 5, which is a prime number,

Order of each element of the group must divide '5'

∴ Order of an element of the above group can be '1' or '5'

• Order can be '1' only for identity element, $\implies 0(0) = 1$

&
for every non-identity element order must be '5' $\implies$

$$0(1) = 5$$

$$0(2) = 5$$

$$0(3) = 5$$

$$0(4) = 5$$



2 mins Summary



✓ Topic

Finite group

✓ Topic

Addition modulo 'm' $\oplus_m$

✓ Topic

Multiplication modulo 'm' $\otimes_m$

✓ Topic

Order of an element in a group $(G, *)$

Topic

Subgroup

THANK - YOU